UNI verification for the log-growth roof, and the joint mixed-LLT operator

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Abstract

On the suspension of the induced Gauss-type map G by the log-growth roof τ, we (i) verify the uniform-non-integrability (UNI) condition for τ — rigorously, from an exact non-proportionality of the branch data — and (ii) set up the joint transfer operator whose local limit theorem yields the native-resolution equidistribution that the density form requires. UNI is the hypothesis Dolgopyat's theorem needs; with it verified, the uniform spectral gap for the additive roof follows from settled theory. We are explicit that a tempting shortcut (cohomology of τ to the geometric potential) is *false*, and that the uniform gap and the mixed LLT are then invoked, not re-proved.

1 The model roof

In the scale-invariant model, a section state is (a, b) with $\xi = b/a$ and offset $o = b - a = a(\xi - 1)$. A return in branch k applies the integer matrix $P_k = \left(\begin{smallmatrix} p_k & q_k \\ r_k & s_k \end{smallmatrix}\right) = C_\ell^k F_\ell$, so $a' = p_k a + q_k b$, $b' = r_k a + s_k b$, and $o' = b' - a' = (r_k - p_k)a + (s_k - q_k)b = a\, L_k(\xi)$ with

$$L_k(\xi) := (r_k - p_k) + (s_k - q_k)\,\xi.$$

Hence the log-growth roof is, exactly,

$$\tau(\xi, k) = \log \left| \frac{o'}{o} \right| = \log |L_k(\xi)| - \log |\xi - 1|. \tag{1}$$

The term $-\log|\xi - 1|$ is branch-independent, a function of the source alone; it is a continuous coboundary and drops out of every temporal-distance comparison.

Remark 1 (a shortcut that fails). One might hope τ is cohomologous to $-\frac{1}{2}\log|G'| +$(locally constant), reducing UNI to the geometric potential. It is *not*: the identity $\tau \stackrel{?}{=} -\frac{1}{2}\log|G'| + \frac{1}{2}(k+1)\log 3 + (\log z' - \log z)$ fails numerically (mean discrepancy 1.32, correlation 0.57 over 1.3×10^6 returns). τ must be treated as its own roof.

2 UNI, verified

Proposition 1 (Non-degenerate temporal distance)**.** *For every pair of distinct branches $i \neq j$ the temporal-distance function*

$$\Delta_{ij}(\xi) := \big(\tau(\xi, i) - \tau(\xi, j)\big) = \log|L_i(\xi)| - \log|L_j(\xi)|$$

is non-constant; equivalently τ is not cohomologous to a function constant on branches. Consequently τ satisfies the uniform non-integrability condition.

Proof. By (1) the common $-\log|\xi-1|$ cancels, leaving $\Delta_{ij}=\log|L_i|-\log|L_j|$. The bottom rows of $P_k=C_\ell^k F_\ell$ give $(r_k-p_k,\,s_k-q_k)=(-4,1),(7,-4),(-19,7),(40,-19),(-97,40),(217,-97),\dots$, whose slope ratios $(s_k-q_k)/(r_k-p_k)=-\frac{1}{4},-\frac{4}{7},-\frac{7}{19},-\frac{19}{40},\dots$ (the $-\rho_k$) are pairwise distinct. Thus L_i, L_j are non-proportional affine forms, so $\Delta'_{ij}(\xi)=\frac{s_i-q_i}{L_i(\xi)}-\frac{s_j-q_j}{L_j(\xi)}\not\equiv 0$ (a nonzero rational function; e.g. Δ'_{02} takes $0.9,-4.5,0.5625$ at $\xi=2,3,5$). Non-constancy of every Δ_{ij} is exactly the statement that no single continuous φ and constants c_k solve $\tau(\cdot,k)=\varphi\circ G_k-\varphi+c_k$; since the branches are real-analytic (Möbius), non-cohomology to locally constant is the UNI condition in the sense of Dolgopyat/Naud/Stoyanov. □

Proposition 2 (Uniform gap, from UNI)**.** *G is a real-analytic full-branch expanding Markov map with a spectral gap, and τ is a real-analytic roof satisfying UNI (Prop. 1) and non-lattice (its increment law has characteristic function bounded away from 1 off the origin, verified). By Dolgopyat's theorem for such systems (cited: Dolgopyat 1998; Naud 2005; Stoyanov 2011), the twisted operator $\mathcal{L}_{i\xi}f=\sum_k |g_k'|^{s^*}\mathbf{e}(\xi\tau_k)f(g_k\cdot)$ has spectral radius bounded away from 1 uniformly for $|\xi|\ge\varepsilon$: $\sup_{|\xi|\ge\varepsilon}\|\mathcal{L}_{i\xi}^n\|\le C\theta_\varepsilon^n$, $\theta_\varepsilon<1$.*

This is the additive-roof gap that was unavailable for the multiplicative cost; it is invoked here, its hypotheses (UNI, non-lattice, analytic gap at $\xi=0$) now discharged.

3 The joint operator and the local limit theorem

Let $\mathcal{L}_{i\xi}$ act on $BV[0,1]$ as above. Its leading eigenvalue $\lambda(i\xi)$ defines the pressure $P(i\xi)=\log\lambda(i\xi)$, analytic near 0 with

$$P(0)=0,\qquad P'(0)=i\Lambda,\qquad P''(0)=-\sigma^2,\quad \Lambda\approx 0.51,\ \sigma^2\approx 0.265>0.$$

The pair $(z_d,\,\ell_d)$, $\ell_d=\log \mathrm{scale}_d=\sum_{j<d}\tau(z_j)$, is governed by $\mathcal{L}_{i\xi}$: for $\varphi\in BV$, $\mathbb{E}_z[\mathbf{e}(\xi\ell_d)\,\psi(z_d)]=\langle\psi,\mathcal{L}_{i\xi}^d\varphi\rangle$.

Theorem 1 (Mixed LLT; cited assembly)**.** *Under Prop. 2 and $\sigma^2>0$, the Nagaev–Guivarc'h method (Aaronson–Denker for Gibbs–Markov maps) gives a local limit theorem: for bounded intervals J and $\psi\in BV$,*

$$\sqrt{d}\;\mathbb{E}_z\big[\mathbf{1}_{\{\ell_d-\Lambda d\in J\}}\,\psi(z_d)\big]\;\longrightarrow\;\frac{|J|}{\sqrt{2\pi\sigma^2}}\int\psi\,d\mu,$$

uniformly in the centering, and jointly with the equidistribution of z_d w.r.t. μ. In particular $\ell_d \bmod \log M$ equidistributes, jointly with z_d, at a polynomial-in-M rate.

4 From the joint LLT to native-resolution equidistribution

At a crossing of scale S, the offset is $o_d=z_d\,e^{\ell_d}$; the fatal event $o_d\in\{1,3,5\}$ is, to resolution $M\sim S$, a joint condition on ($z_d \bmod$ -, $\ell_d \bmod \log M$). Theorem 1 supplies exactly the joint equidistribution needed: the per-crossing hazard is $6h(0)/S+o(1/S)$ with the error controlled by the LLT rate, uniformly in S. This is the content of the open uniform-gap lemma ($\star$), now delivered through the additive-roof route.

5 Status

Rigorous here: the model roof (1); the UNI condition for τ via the exact non-proportionality of the branch forms (Prop. 1); the pressure data $P(0) = 0$, $\sigma^2 > 0$; the falsity of the geometric-potential shortcut. *Invoked (settled theory, hypotheses discharged):* the uniform gap from UNI (Dolgopyat/Naud/Stoyanov, Prop. 2); the mixed LLT (Nagaev–Guivarc'h / Aaronson–Denker, Thm. 1). *Still to write:* the $O(1/S)$ model-to-exact corrections carried through the LLT (via the truncation of the formal supplement); and the fully detailed Nagaev–Guivarc'h perturbation for the joint operator.

The open lemma is therefore no longer "outside the theory needing a new idea." Its one non-standard hypothesis — UNI for a genuinely multiplicative-looking cost — has been reduced, via the suspension, to UNI for an additive real-analytic roof, and *that* is now verified from the exact branch arithmetic. What remains is the invocation of two standard theorems whose hypotheses are discharged, plus routine perturbation bookkeeping. No closure is claimed, but the analytic obstruction is removed.